

$$\Theta_1 = 0.652, \Theta_0 = 0.766$$

$$X = [1, 2, 3], Y = [3, 5, 7]$$

$$\eta = 0.01, N = 3$$

Step 1

Prediction

$$\hat{y}_1 = 0.766 + 0.652 \times 1 = 1.418$$

$$\hat{y}_2 = 0.766 + 0.652 \times 2 = 2.07$$

$$\hat{y}_3 = 0.766 + 0.652 \times 3 = 2.722$$

Step 2 = Errors

$$e_1 = \hat{y}_1 - 3 = -1.582$$

$$e_2 = \hat{y}_2 - 5 = -2.93$$

$$e_3 = \hat{y}_3 - 7 = -4.278$$

Step 3 = MSE

$$L = (1/N) \sum e_i^2$$

$$e_1^2 = 2.502724$$

$$e_2^2 = 8.5849$$

$$e_3^2 = 18.301284$$

$$\text{Sum} = 29.38891$$

$$L = \text{Sum}/3 = 9.796303$$

Step 4 Gradient w.r.t θ_1

$$x_1 e_1 = 1 \times e_1 = -1.582$$

$$x_2 e_2 = 2 \times e_2 = -5.86$$

$$x_3 e_3 = 3 \times e_3 = -12.834$$

$$\text{Sum} = x_1 e_1 + x_2 e_2 + x_3 e_3 = -20.276$$

$$\partial L / \partial \theta_1 = (2/3) \times \text{Sum} = -13.5173$$

Step 5 Gradient w.r.t θ_0

$$\partial L = \partial \theta_0 = (2/N) \sum e_i$$

$$\text{Sum} = e_1 + e_2 + e_3 = -8.79$$

$$\partial L / \partial \theta_0 = (2/3) \times \text{Sum} = -5.86$$

Step 6 update

$$\theta \leftarrow \theta - \eta \times \text{Gradient}$$

$$\theta_1 - \text{new} = 0.652 - 0.01 \times (\partial L / \partial \theta_1)$$

$$= 0.787173$$

$$\theta_0 - \text{new} = 0.766 - 0.01 \times (\partial L / \partial \theta_0)$$

$$= 0.8246$$